

1. What is an Operating System?

An Operating System (OS) is software that manages computer hardware resources and provides common services for computer programs. It acts as an intermediary between user applications and the underlying physical hardware.

2. What is a kernel?

The kernel is the core component of an operating system. It operates at the lowest level, directly interacting with hardware to manage critical functions such as system memory, CPU scheduling, input/output operations, and device drivers.

3. Which other parts aside from the kernel identify an OS?

Aside from the kernel, an OS includes:

System Libraries: Standard code routines that applications use to interact with the kernel.

System Utilities & Shell: Command-line interfaces (CLIs) or user environments that allow users and scripts to issue commands to the system.

Graphical User Interface (GUI): Desktop environments (e.g., GNOME, KDE) for visual interaction.

Core Application Suites & Package Managers: Software management tools and essential software necessary for standard system tasks.

4. What is Linux?

Technically, Linux is a free, open-source monolithic kernel originally created by Linus Torvalds in 1991. In common usage, "Linux" refers to the complete Unix-like operating system that combines the Linux kernel with GNU utilities, software libraries, and additional system applications.

5. What is a Linux distribution?

A Linux distribution (or "distro") is an operating system built from the Linux kernel, system utilities, management tools, software packages, and a desktop environment. Organizations or communities bundle these components together into an installable operating system tailored for specific use cases (e.g., desktop, server, security testing).

6. List at least 4 Linux characteristics:

Open Source: The source code is freely available for anyone to view, modify, and redistribute.

Multi-user & Multitasking: Allows multiple users to access system resources simultaneously and run multiple processes concurrently without interference.

High Security & Stability: Strong permission models, process isolation, and robust network security features make it stable and less vulnerable to malware.

Portability: Can run on a wide variety of hardware architectures, from supercomputers and servers to embedded devices and personal computers.

7. What is Debian?

Debian is one of the oldest, most stable, and influential open-source Linux distributions. Maintained by a global community of volunteers, it relies on the dpkg and apt package management systems and serves as the upstream foundation for many popular distributions, including Ubuntu and Kali Linux.

8. List and define the different types of licensing agreements

Permissive Licenses (e.g., MIT, Apache, BSD): Allow broad freedom to use, modify, and redistribute code, including within proprietary/commercial software, with minimal restrictions beyond maintaining copyright notices.

Copyleft / Protective Licenses (e.g., GNU GPL): Require any derivative works or modified versions of the software to also be released under the exact same open-source license.

Proprietary Licenses: Restrict access to the source code, prohibit redistribution or modification, and grant users only specific, limited rights to execute the compiled software.

Public Domain: Software with no copyright restrictions whatsoever; anyone can use, modify, or sell it without conditions.

9. What is Free Software? Define the 4 freedoms.

Free Software (as defined by the Free Software Foundation) refers to software that respects users' freedom and community, giving them the control to run, adapt, share, and improve it. "Free" refers to freedom ("liberty"), not price.

The 4 Essential Freedoms (numbered 0 through 3):

Freedom 0: The freedom to run the program for any purpose.

Freedom 1: The freedom to study how the program works and change it so it does your computing as you wish (access to source code is a prerequisite).

Freedom 2: The freedom to redistribute copies so you can help your neighbor.

Freedom 3: The freedom to distribute copies of your modified versions to others, giving the whole community a chance to benefit from your changes.

10. What is virtualization?

Virtualization is the process of creating a software-based (virtual) representation of something—such as virtual hardware, storage, or network devices—rather than a physical one. It enables a single physical machine (host) to run multiple isolated operating systems (guest virtual machines) simultaneously using a layer called a hypervisor.